Simple ATM Deposit and Withdrawal

1. Problem Statement

Create a simple ATM program that allows a user to deposit money or withdraw money from an initial account balance. Display the updated balance.

2. Objectives

1. Store the account balance in a variable.
2. Read transaction type and amount.
3. Use conditions to process deposits and withdrawals.
4. Display the updated balance.

3. Assumptions

1. Initial balance is ₹5,000.
2. Transaction amount must be positive.
3. Withdrawal is allowed only when sufficient balance is available.

4. Algorithm

1. Start.
2. Set balance to 5,000.
3. Read transaction type and amount.
4. If transaction is Deposit, add amount to balance.
5. If transaction is Withdraw, check balance and subtract amount if sufficient.
6. Otherwise, display an invalid transaction message.
7. Display final balance.
8. Stop.

5. Flowchart

None

6. Pseudocode

```
None
balance = 5000

INPUT choice, amount

IF amount <= 0 THEN
    DISPLAY "Invalid amount"

ELSE IF choice = "deposit" THEN
    balance = balance + amount

ELSE IF choice = "withdraw" AND amount <= balance THEN
    balance = balance - amount

ELSE
    DISPLAY "Invalid transaction"

ENDIF

DISPLAY balance
```

8. Test Case and Expected Output

Input:

```
None
Initial balance = Rs. 5000
Choice = withdraw
Amount = Rs. 1200
```

Calculation:

Updated Balance = Initial Balance − Withdrawal Amount

Updated Balance = ₹5,000 − ₹1,200 = ₹3,800

Expected Output:

```
None
Initial balance: Rs. 5000
```

```
Enter deposit or withdraw: withdraw
Enter amount: Rs. 1200
Available balance: Rs. 3800.0
```

9. Conclusion

The program simulates a basic ATM transaction using variables and conditional statements. It allows users to deposit money or withdraw money when sufficient balance is available and displays the updated account balance.